

Junwon Seo

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RESEARCH INTERESTS

Methodology	Active learning · Anomaly detection · Time-series forecasting · Combinatorial optimization
Application	Quality assurance in manufacturing (semiconductor wafer maps, additive manufacturing) · Renewable energy forecasting

EDUCATION

Hanyang University, Seoul, Korea	Mar 2020 – Feb 2027 (expected)
• B.S. in Industrial Engineering (primary major) • B.S. in Mathematics (double major) • Micro-Major in Semiconductor	
Mandatory military service, Republic of Korea Army, Mar 2022 – Sep 2023	

RELEVANT COURSEWORK

Industrial Engineering	Applied Data Analytics · Smart Manufacturing Data Analytics · Time Series Analysis and Forecasting · Design of Experiments · Computational Statistics · Applied Statistics · Probability and Statistics · Numerical Analysis · Linear Programming · Management Science and Operations Research II · Capstone PBL: Reinforcement Learning Theory and Applications · Artificial Intelligence and Machine Learning · AI+X: R/Python Computing · Data Structures · Object-Oriented Programming
Mathematics	Real Analysis · Probability Theory · Mathematical Statistics I · Linear Algebra I · Advanced Calculus I, II · Modern Algebra I, II · Complex Analysis I · Combinatorics · Differential Equations · Mathematical Programming · Computational Mathematics
Semiconductor	Plasma Equipment · Introduction to Electrical Energy

RESEARCH EXPERIENCE

Undergraduate Research Student, Smart Manufacturing and Data Analytics (SMDA) Lab	Mar 2026 – Aug 2026
<i>Hanyang University · Advisor: Prof. Jihoon Chung</i>	

Label-efficient defect detection with active learning

- Studied semi-supervised label spreading and active learning on MNIST, then reproduced a mixed-supervision surface-defect segmentation model (Božič et al., 2021) on KSDD2 using the authors' public code.
- Replaced the paper's random choice of which 16 defect images receive pixel-level labels with active-learning strategies; core-set (k-center) selection reached AP 0.881 versus 0.865 for random selection.
- Extended the study to semiconductor wafer maps (WM-811K), which have no die-level ground truth: generated 20,000+ defect masks by running 10 Claude Opus agents in parallel on an expert-style labeling procedure, and validated them on 846 defect-free wafers (median false-marking rate 0–2.1%).
- Over 10 runs with 256 labels, k-center improved IoU 1.48× over random selection (0.321 vs. 0.217, $p < 0.001$) and captured rare defect types ($< 1\%$) that random selection missed; uncertainty sampling lost to random in all 10 runs.
- Worked through the proofs of the core-set bound (Sener & Savarese, 2018) and confirmed that the measured coverage radius δ_S ranked the strategies in the same order as their performance (k-center 3.8 vs. random 16.0).

Undergraduate Thesis, Industrial Engineering Capstone Design I	Mar 2025 – Jun 2025
<i>Hanyang University · Team of 3</i>	

Anomaly detection for a thermal power plant using sensor data

- Handled preprocessing and LSTM-autoencoder modeling on 1,108 sensor variables recorded every minute (4,320 time points).
- Removed 202 constant sensors and a distribution-shifted day (Kolmogorov–Smirnov test), clustered sensors with DBSCAN, and kept the five highest-variability sensors per cluster as 50 model inputs.
- Reconstruction-error detection reached precision 0.99, recall 0.97, and F1 0.98, compared with F1 0.80 for a Hurst–Mahalanobis control chart.

HONORS & AWARDS

Excellence Award, 8th of 985 teams (top 0.8%)

Sep 2026

BARAM 2026: 3rd Wind Power Generation Forecasting AI Competition · Korea East-West Power, GS E&R, Taebaek Gadeoksan Wind Power, DACON · Team leader (3 members)

- Forecast hourly output of three wind-farm groups (17 turbines) over 8,760 hours with 38 LightGBM models built on physics-based features such as a SCADA power curve and air density × wind speed³.
- Added reverse-time validation (train on 2024, validate on 2023) and adopted only changes that improved both directions; the public rank stalled at 59th, and the final private rank was 8th.

Top Excellence Award, 3rd place

May 2026

24th University Student Project Competition · Korean Society of Industrial and Systems Engineering (KSIE) · Team of 5, data collection and model design

Real-time anomaly detection for additive manufacturing based on the Laplacian spectrum

- Built a 3σ control chart on λ₂ of the normalized graph Laplacian of 6-axis vibration signals, using only three normal prints as the baseline.
- Detected an internal-pattern defect in 20 s (LSTM autoencoder 63 s, one-class SVM 65 s) and was the only method to catch an internal void defect; implemented as a real-time FastAPI/WebSocket dashboard.

COMPETITIONS

LG CNS Optimization Grand Challenge 2026 (global), 43rd of 958 teams (top 4.5%)

Jun 2026 – Aug 2026

LG CNS · Korean Institute of Industrial Engineers (KIIE) · Team leader

- Shipyards 3D block placement and scheduling. Built a C++ placement engine (NFP, LBB, ALNS); a rule that deliberately delays the largest block cut the objective by up to 31% on heavy-tardiness instances.
- Trained a placement-scoring policy with reinforcement learning on 600 instances augmented from 20, reducing the objective by up to 23.8% on mixed instances.

NYPC Code Battle 2025, top 14.3% (online round, performance 2120)

Sep 2025

Nexon · Game-playing strategy bot in Python

LG CNS Optimization Grand Challenge 2025, 67th (preliminary round)

Jul 2025

Korean Institute of Industrial Engineers (KIIE) · LG CNS

ACTIVITIES & TEACHING

FIELD (National Association of Industrial Engineering Students), 18th Cohort, Competition Division

Feb 2026 – Aug 2026

- Set problems for FIELD CAMP (300 participants), writing the AI data-center siting optimization part; proposed the camp’s first essay-type problem graded on reasoning and wrote its scoring rubric. Named Outstanding Member (first half of 2026).

Mathematics Teaching Assistant, Daechi Russell (private academy), Seoul

Jan 2024 – Present

- Answer questions, grade work, and track each student’s recurring errors for high-school seniors and repeat test-takers preparing for the college entrance exam.

CERTIFICATIONS & SKILLS

Certifications	Advanced Data Analytics Semi-Professional (ADsP), K-DATA, Jun 2026 · SQL Developer (SQLD), K-DATA, Sep 2026 · Six Sigma Green Belt (Level 1), Aug 2026
Languages	Korean (native) · English (TOEIC Speaking 140, Intermediate High, Aug 2026)
Programming	Python · C++ · Java · R · SQL · MATLAB
Libraries & Tools	PyTorch · LightGBM · Gurobi · CP-SAT · FastAPI · WebSocket · MQTT · Arduino · Claude Code · Codex